

Gendering Robots in Human-Robot Interaction: An Interdisciplinary Approach

Neziha Akalin

*Dept. of Computer Science and Informatics
School of Engineering
Jönköping University
Jönköping, Sweden
neziha.akalin@ju.se*

Maria Arnelid

*Dept. of Thematic Studies
Unit of Gender Studies
Linköping University
Linköping, Sweden
maria.arnelid@liu.se*

Katherine Harrison

*Dept. of Thematic Studies
Unit of Gender Studies
Linköping University
Linköping, Sweden
katherine.harrison@liu.se*

Abstract—This paper emphasizes the importance of interdisciplinary collaboration in exploring the concept of robot gender within Human-Robot Interaction (HRI). It draws on a case study of the authors' own collaboration, where interdisciplinary discussions on the nature of gender informed the design of a multi-method study. This approach allowed us to avoid assigning a binary gender to Pepper robot, while still encouraging participants to reflect on their own gendering practices during interactions with robots. Additionally, we invite readers to consider alternative ways of conceptualizing robot gender in HRI—specifically, as fluid and performative rather than binary, which relies on stereotypical cues. After describing how our discussions on gender influenced and reshaped the study design, we offer practical advice on fostering interdisciplinary collaborations. These suggestions focus on communication strategies, mindset, and the practical setup of collaborative studies. We hope that these recommendations will inspire other researchers to continue exploring new and interdisciplinary ways of approaching robot gender in HRI.

Index Terms—Gender, Robots, Interdisciplinary communication, Human-robot interaction.

I. INTRODUCTION

This paper calls for interdisciplinary approaches to gender in HRI. In recent years attention has turned to the role played by gender in human interactions with social robots [1]–[8]. These works acknowledge the importance of gender within social interactions. They often investigate how robots may be gendered to facilitate better interactions with a view to long term, widespread deployment and adoption in various sectors of society such as healthcare or education. Existing HRI literature often presents gender as binary and rooted in stereotypical cues, which contrasts with gender studies research that views gender as fluid and performative, behaving in particular ways to fit into society. Many of the above rely on a relatively simple model of gender which focuses on expressing gender through one or two gender cues, such as dressing [9], body type [10], voice [11], [12] or manipulation of names and pronouns [5]. This paper details an interdisciplinary experiment that explores how a more nuanced understanding of gender might be developed and employed to improve HRI.

An interdisciplinary approach integrates different perspectives, theories and techniques to address complex societal

problems. It promotes creativity and critical thinking however, it can also be challenging as participants must bridge disciplinary gaps to collaborate effectively. Although literature on interdisciplinary work is widely available [13]–[16], this paper highlights some of the specific challenges encountered in an HRI context, and specifically focus on those related to the robotic body. We share insights from our own learning process and provide guidelines for researchers interested in pursuing similar research. The case study discussed in this paper stems from an interdisciplinary collaboration between the authors—who have backgrounds in Gender Studies and HRI. This collaboration revealed both the benefits and challenges of applying interdisciplinary methods to studying gender in HRI.

The authors of this paper are all researching various aspects of the development of social robots for care environments, in different disciplinary contexts. The project aims to conduct interdisciplinary research on caring robots, with a focus on trust, empathy, and accountability. As part of this broader initiative, we decided to design a user study centered around trust and robots in healthcare settings.

A. Video-based Study: Robots in Healthcare

We designed a video-based between-subjects user study using the humanoid robot Pepper to investigate how assigning different professional roles to the robot in a healthcare setting affects participants' trust in the robot. Three videos were recorded in which Pepper took on different care provider roles: an orthopedic specialist, a nurse, or a physiotherapist, corresponding to the study's three experimental conditions. A snippet from the video in which Pepper took a physiotherapist role is given in Figure 1.

In the study, each participant was assigned to one of the three conditions. After watching the video, they completed a questionnaire assessing their level of trust in the robot. Further, participants were given the opportunity to participate in a follow-up interview. In the interviews, participants were asked in-depth questions about their thoughts about Pepper in particular and about robots in healthcare in general.

Although gender was not explicitly incorporated into the study design, it was highly present throughout the development of the study. All authors agreed that gender was a relevant

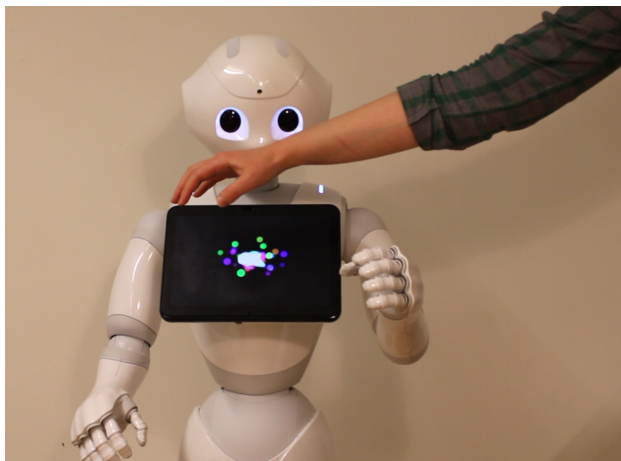


Fig. 1. A snippet from the video-based study in which Pepper examines the patient's wrist in physiotherapist role.

factor that might shape trust in care providers, yet we struggled to find the proper place for gender in the research design. Our challenge became how to explore the impact of a robot's perceived gender in a way that fulfilled both our disciplinary understandings of gender as well as the affordances and limitations of gendering a robot body. As we experimented with different approaches, the role of gender evolved throughout the study's design process. We believe that shifting this role and the interdisciplinary discussions that accompanied it, offer valuable insights for studying gender in HRI. Below, we will explain the process of finding the role of gender in our research design.

II. DEVELOPING THE EXPERIMENT

In this section, we detail the development of the interdisciplinary discussions, and how these discussions shaped the user study.

A. Initial discussions

Our decision to collaborate followed an initial meeting between the two research teams. After identifying a shared interest in trust, robots and healthcare, we agreed to pursue a collaborative study. We discovered that we were all interested in how the professional role of a robotic care provider affects trust, and decided to center our study around this topic.

During these discussions, it became evident that we all believed gender could influence the relationship between professional role and trust. Following similar lines of thought, research in HRI has investigated the relationship between gender-related occupational stereotypes and trust [10], [17], [18]. While some studies show minimal impact of robot gender on trust in the robot's capabilities [17], others show that participants preferred a robot whose gender-occupational role and personality-occupational role stereotypes matched [18]. After establishing that we were all interested in contributing to the discussions being had regarding the relationship between gender, occupational role and trust in HRI, our discussions

quickly turned into questions of how to incorporate this dimension into the experiment in a practical way.

B. Learning Seminar

Communication between different disciplines can be challenging due to varying terminologies which may lead to difficulties in aligning perspectives. For example, what is considered fundamental knowledge in one discipline may be unfamiliar in another, which can create gaps in understanding or slow down collaborative efforts. To this end, a learning seminar can be highly beneficial in addressing these issues, providing a structured space for participants to share their disciplinary perspectives, clarify concepts, and build a common language. Before starting to discuss the details of experimental design, we arranged a learning seminar which significantly improved our communication and enabled more effective collaboration. During the seminar, the Gender Studies scholars presented the concept of gender as a performance while the HRI scholar presented the robot and the Trust-Perception Scale [19] and its potential use for the experiment.

C. Gender as variable

After the learning seminar, our discussions focused on how to include gender as a variable in the experiment using gender cues. Since we wanted to investigate the role gender might play in trust relations, we planned to record two versions of each video scenario (Pepper as an orthopedic specialist, Pepper as a nurse, and Pepper as a physiotherapist). One version would portray Pepper with traits associated with men, and the other with traits associated with women. Following this initial idea, we discussed different ways to incorporate gender cues into the study.

First, we addressed the possibility of assigning Pepper a gender by altering its appearance and studying the differences in responses accordingly. Two examples of ways in which robot appearance have been gendered in previous HRI research, that we discussed, is by altering the robot's hairstyle [9] and giving the robot different accessories. The latter has been done for example by dressing it in a pink bow that is placed either on the top of its head or on its neck (as a blue bow tie) [11]. These options illustrate the limited range of options for visibly gendering the robot. They reflect the material limitations of the robot (Pepper has a white plastic casing with static features, unlike other humanoid robots such as Furhat).

In discussing these options, we shared concerns that gendering a robot through visual cues like hairstyle and accessories might risk reproducing harmful gender stereotypes about what women and men look like [7]–[10]. We also felt a tension between displaying a robot's gender through such cues and the ongoing discussions being had in Gender Studies. More specifically, discussions regarding gender as a socio-cultural and context specific performance, comprising voice, dress, movement and other aspects [20], [21].

Following this concern, we turned to the option of incorporating gender cues by altering Pepper's voice settings. We considered altering between a voice coded to sound like a

woman and one coded to sound like a man. We also discussed changing Pepper's name to one typically associated with men and another typically associated with women.

D. Scripting gender

Around the time we were considering using voices and names as gender cues, we began drafting a script to guide the dialogue between Pepper and an actress playing the role of a patient in the videos. We decided to use the script-writing process as an opportunity to test these gender cues, writing scripts in which Pepper's gender would be conveyed by altering its voice and name. This involved creating two versions of the script for each professional role (nurse, orthopedic specialist, physiotherapist): one where Pepper was gendered as a woman and one where Pepper was gendered as a man.

When we started detailing how Pepper would be gendered in the script for the videos, we realized that our earlier discomfort with using appearance-based gender cues still persisted. On further discussion, we realized that what bothered us was the practice of assigning gender *onto* the robot. In other words, we found that the options for gendering a robot body resembled imposing simple, stereotypical notions of gender onto an ungendered—without considering the importance of context or interaction. This approach lacked reference to the understanding of gender common in Gender Studies, which we mentioned briefly earlier. In this understanding, gender is seen as a daily performance that must be constantly done and re-done to render the lived body visible and recognizable to others. This model also implies context, interaction, a multiplicity of cues and a constant struggle between normative demands for coherent recognizable gender expression and individual expressions of one's body. In essence, when humans “do” gender [22], this involves agency, nuance and complexity.

In these interdisciplinary discussions, several key questions emerged. How can we translate this understanding of gender into the context of a robotic body? How can we study the impact of robot gender in HRI without relying on stereotypes? These questions highlight a disciplinary and methodological tension. On one hand, we want to investigate the role of gender in shaping trust in HRI, aiming for a clear answer as to whether trust improves when the robot is perceived as a “man” or a “woman”. On the other hand, there is the discourse in gender studies, which views gender as fluid, performed or done—happening in interactions between different bodies.

This tension led us to ask: instead of imposing human gender norms onto Pepper, what could robot gender look like? In other words, perhaps we need a new way of conceptualizing robot gender, rather than directly mapping human gender categories onto robots. Given the limitations and affordances of the robot body, what would constitute a performance of a robot's gender identity? [23], [24].

E. (Re)moving gender

Upon encountering the tension between using gender cues in HRI and the discussions in Gender Studies, we decided to “move” gender in the experiment. This meant that gender

would no longer be included explicitly as a variable, and we would not assign gender cues to the robot in the videos. By doing so, we avoided using explicit gender markers (such as bows or voice) and could instead approach gender in other ways.

While removed gender as a variable, we were still interested in understanding its role in the relationship between trust and professional roles in HRI. To address this, we shifted our focus from creating gender on the robot's body to observing how gender emerged in the interaction itself. We made two changes to our research design:

- We added a multiple choice question to survey, asking “Did you perceive Pepper as having a gender?” Participants were given the following options to answer the question: ‘Yes’, ‘No’ and ‘Not sure’. Those who answered No or Not sure were directed to the next question in the survey. Those who answered ‘Yes’, however, were directed to a follow-up question asking “Could you say a bit more?”. The follow-up question was added to encourage participants to reflect on their gendering practices already in the survey. We made the choice of keeping the follow-up question optional and free-text to invite a wide range of answers regarding participants' gendering practices. Further, we chose to phrase the question in a fairly open way (as opposed to asking, for example, “which gender did you think of Pepper as?”), to invite participants to reflect on gender in ways that are not necessarily binary (man/woman).
- We included qualitative follow-up interviews as part of the study design, which also allowed for deeper discussions on trust in robots. Participants who agreed to be contacted for an interview provided their email at the end of the survey. These interviews, lasting 30-45 minutes, included questions on both trust and gender. When gender was mentioned in the survey, participants were asked to reflect on their responses. In some cases, participants spontaneously used gendered pronouns (he/she) when referring to Pepper, and were asked to explain their choice. These discussions helped us explore the nuances of gendering practices and allowed participants to reflect more deeply on their perceptions of robot gender in the interaction.

III. CARRYING OUT THE EXPERIMENT

We have outlined how gender's role in our research design evolved—shifting from being used as a variable to being brought up in the survey in a subtle way and in follow-up interviews. We also discussed the interdisciplinary conversations about how to approach and study gender, which led us to this decision. Now, we will explain what moving of gender meant for our final experiment setup. The decision influenced the structure of the three video conditions in several ways. We chose not to modify Pepper's physical appearance and used the default, neutral voice setting. In other words, we did not actively assign a gender to Pepper's physical being.

The decision also impacted the content of the script. One of Pepper's first lines mentions its name. Initially, the plan was to record two versions of each scenario—one where Pepper was assigned a voice and name associated with women, and another where Pepper was assigned a voice and name associated with men. However, after removing explicit gender cues, we recorded only one version of each video condition. In the final version, Pepper used the neutral voice setting and introduced itself simply as "Pepper". The only difference in Pepper's introductions across the three video conditions was the professional title: nurse, orthopedic specialist, or physiotherapist. While the script was slightly adapted for the different roles, there were no adjustments made to assign Pepper any specific gender. In other words, no explicit gender was assigned to Pepper in the script.

The three different conditions were video-taped, edited and included in a survey platform together with a survey consisting of two parts¹. The first part was the Trust-Perception Scale [19] which consists of a series of percentage questions pertaining to trust. The second part was added to include more context-sensitive questions about what participants found important about a robotic healthcare provider. This is also the section where we included the question "Did you think of the robot as having a gender?". The last question in the survey asked whether the participant would be interested in taking part in a follow-up interview, and if so they were asked to fill out their email address. The interviews are semi-structured, meaning that they are partly based on pre-defined questions but also allow for the interviewer to steer the interviews partly according to the interlocutors' responses [25]. The questions had to do with participants' thoughts on trust and gender, as well as other general impressions that they had of Pepper in the video. The interviews range from 30-40 minutes and, so far, have all been recorded after obtaining the written consent by the participants.

A. Participants

To date, we have received 68 survey responses and conducted seven follow-up interviews. Out of 68 participants, 29 identified themselves as men, 37 as women, and 2 as non-binary. The largest age group was between 25-34 years old (34 participants), followed by 35-44 years (12). Nine participants were 65 and over, six were aged 55-64, five were in the 45-54 age group, and two participants were between 18-24. In terms of educational background, 32 participants held a Master's degree, 16 had a Doctorate, 14 had a Bachelor's degree, four had a high school degree or equivalent, and two had vocational training. The majority of participants were based in Sweden (33), however, we had participants from Turkey (3), the United Kingdom (6), the United States (4), Germany (3), Canada (3), Australia (2) and participants from 11 other countries. This diverse group provided a wide range of perspectives for the study.

¹The video based study can be reached here: <https://bit.ly/3Ejf3ws>

IV. RESULTS

We are currently exploring how the qualitative and quantitative data can complement and enrich one another. The outcomes of these analyses will be presented in future publications.

Twelve participants answered "Yes" to the question, "Did you think of the robot as having a gender?" Of these, 5 of them saw the video where the robot acted as an orthopedic specialist, 1 saw the robot as a physiotherapist, and 6 saw the robot as a nurse. Ten participants was "Unsure" whether the robot has a gender, making a total of 22 out of 68 participants.

This indicates that even after we removed explicit gender cues, nearly 30 percent of participants still perceived the robot as having a gender or were uncertain about it. When asked to explain why they thought the robot had a gender, only 3 out of the 12 thought the robot was male, while the majority thought it was female, due to various reasons. Their responses highlighted the different cues which made participants gender the robot such as a mix of voice, appearance, name, and internalized cultural roles to assign gender to the robot.

Twelve participants who answered "Yes" also shared their reasoning for the robot as having a gender, despite the absence of explicit gender cues. Some attributed their perceptions to the robot's voice, such as one participant said, "The voice sounded more female" while another noted, "The voice gave me this thought" and described it as a "female-sounding voice". Another participant found the voice "annoying" and referred to it as a "fake female voice."

Others linked gender perception to a combination of visual cues and roles. One participant remarked, "Body shape and role (nurse) both implied, in the Australian context, a female role", while another added, "Given the voice and appearance, as well as internalized gender roles, I would think of this robot as being gendered as a woman". The role of the nurse specifically seemed to suggest a female identity for some, with one participant saying, "The robot seemed to be female to me."

Not all participants saw the robot as female. One remarked, "It felt like it was a boy", another stated, "In my mind, the robot is automatically male. I can't explain why." The name "Pepper" also influenced perceptions, with one participant saying, "The name Pepper sounded like a name for a male. But the voice sounded like it was from a female." Despite this, they concluded, "Most of the time, I thought of the robot as having a female gender."

A Mann-Whitney U test was conducted to compare participants' trust scale ratings for Group 1 (the ones who thought the robot had a gender or were unsure) and Group 2 (those who did not perceive the robot as having a gender). The test revealed no significant difference between the two groups ($U = 380$, $p = 0.1$). Participants in Group 1 had a median trust rating of 5.03 and those in Group 2 had a median rating of 5.93. These results suggest that the participants' gender attribution to the robot did not significantly influenced their trust in robot in a given care professional role.

V. DISCUSSION

Our main aim with this paper is to argue for the benefits of an interdisciplinary approach to studying gender in HRI. To do so, we have outlined the process of designing an experiment within an interdisciplinary team. Our interdisciplinary discussions led us to reconsider and conceptualize the role of gender in HRI in new and fruitful ways.

Instead of assigning a binary gender to Pepper, this approach enabled us to engage participants in open-ended discussions about gender. We phrased the question regarding gender carefully which allowed participants to reflect on their own gendering practices. Rather than treating gender as something fixed and binary, we embraced the idea of gender as a dynamic and performative that emerges during interaction. This approach aligns with the broader understanding of gender in Gender Studies, where gender is seen as fluid which is continuously done and undone.

A. Gender Performance and Limitations of Robotic Body

The motivation behind moving gender from the experimental design lays in the limitations of the robotic body. Human beings dress, style and move their physical bodies as part of the ways in which we indicate our gender, or other identifying characteristics. Pepper has limited options in this respect. For example, the facial features are static, and although Pepper's arms move, the fingers are mostly for show. Pepper can bend and turn its torso, but not move from the spot. It can interact with the humans around it through advanced perception and recognition programming. However, Pepper's own options for expression are rather limited.

Gender studies scholars have emphasized that the ways in which humans perform their gender are specific to particular contexts and policed as part of a broader set of sociocultural norms around the "proper body" [26], [27]. When bodies do not fit these norms, they are often more visible or subject to higher levels of scrutiny, or even harassment. Trying to gender Pepper demonstrated how complex these performances are, and how disorienting or distracting an incoherent or poor performance might be within an interaction. How can robots with their limited expressive capacities fit into human sociocultural norms around gender? The answer to this is beyond the scope of this paper, but deserves careful thought as we seek to develop robots that will integrate well into human society.

Further, it is important to mention that "avoiding" gender cues in HRI research design does not "solve" the question of gendering in the context of robots: people may think about robots in gendered terms even without the use of gender cues. The varied responses for the open-ended question suggest that gender is not simply ascribed based on clear-cut cues but emerges through an interplay of voice, role, name, and internalized cultural norms. Humans' robot gendering may be flexible such that the assigned gender may be changed over time due to the task robot performs or the cultural background of people may mediate the gender assignment. That is, the same robot with the same features can be gendered differently

in different cultures. One of the participants mentioned this as they related healthcare roles with females in Australian context.

As noted earlier, there is an existing interest within HRI in the role of gender. More recently, this has developed to considering how ideas about gender and other identity markers shape the design and development of the robotic body, but also the human-robot interaction [28]. Integrating feminist perspectives on the body draws attention to how power structures related to bodies and identities may be (re)produced in human encounters with robots. This is important to consider as it will impact how safe and comfortable humans feel interacting with robots.

The idea of robot gender as performance brings up several ethical questions as this might not fit into social and cultural norms. People may expect particular traits from a robot with a certain role. However, if a robot's gender is based on simple traits like its voice or appearance, it could reinforce stereotypes especially in roles like caregiving or education—it might raise concerns about bias and fairness in human-robot relationships. For example, if robots are gendered in line with traditional roles (like male doctors or female nurses), it could strengthen associations between gender and profession.

B. Lessons from Interdisciplinary Collaboration

Our interdisciplinary collaboration provided several important lessons on how to approach gender in HRI research. First, it is crucial to invest time in providing all collaborators with a broad understanding of each field's perspectives and methodologies. In our case, an early learning seminar helped us bridge key differences in how gender and related concepts are approached in HRI and Gender Studies. Additionally, collaborating on the experimental script allowed us to uncover nuanced differences in our approaches, resulting in a more detailed and thoughtful design.

Second, developing a shared vocabulary is essential. Throughout the project, we frequently discussed our differing views on gender—whether it is something that can be ascribed to a robot or something performed in interaction. These conversations were invaluable in shaping our methodological decisions and ensuring that the study reflected a balanced interdisciplinary approach.

The discussions led us to adopt a multi-method approach, containing both quantitative and qualitative elements. These choices have allowed us to bridge some of the disciplinary differences in between HRI and Gender Studies in how gender is understood and studied. Instead of assigning a binary gender to Pepper, the multi-method approach that we adopted allowed us to engage in discussions with participants about gender. Through carefully phrasing our questions around gender in the survey as well as in the follow-up interviews in open and (hopefully) non-restraining ways, we have attempted to incorporate into an HRI experiment a view of gender as fluid, performative and not binary. With this we mean that it allowed us to avoid *assigning* a binary gender to Pepper, while still allowing participants to reflect on their gendering

practices in interactions with robots. We would therefore like to argue that by moving gender in the research design—from the video to open-ended questions in the survey and follow-up interviews—we also moved gender in another sense. We moved from an understanding of gender as something that can be *shown* and that exists on a binary scale, to an understanding of gender as happening in interaction, as performed, and as fluid.

C. Suggestions for Interdisciplinary Collaborations

The previous section describes the lessons that we learned through our interdisciplinary collaboration around gender in HRI. It also addresses some larger questions about robot gender that came from this work. We would like to encourage other researchers to reflect in interdisciplinary ways around these questions. In order to enable such discussions, we present the following suggestions of how to enable fruitful interdisciplinary collaborations:

- At an early stage of the collaboration, spend time providing each other with a broad background to your field and the key developments relevant to the proposed study. We incorporated this in our study through a learning seminar. This seminar was held early on in the collaboration and provided a space to discuss broadly how our respective research groups relate to common research interests. For example, differences in understanding of *emotions* in emotion recognition [29], [30] versus affect theory [31], [32] were discussed.
- The methodological practicalities of how research is done in the respective fields. For example, research skills such as setting up a robot for an experiment that are familiar to HRI researchers may be unfamiliar to those in the social sciences. Methodological terms that are shared in between disciplines may be performed differently in practice. In our case, developing the experimental script on a shared document together helped us to detect the nuances of the scenario and resulted in a far more detailed script than might have been normal practice in HRI.
- Identify and discuss key concepts that all collaborators share and use, in order to develop a common vocabulary. In this case, a central concept is gender. Throughout our collaboration, we have discussed differences in between thinking of gender as something that is ascribed to a robot versus something that is performed, or done. These conversations are crucial and deserve time. Knowing how all collaborators understand concepts of concern also helps contextualize some of the methodological choices that have to be made.
- Schedule regular “check-ins”—meetings that you have throughout the research project—to avoid misunderstandings and to encourage sharing reflections on the ongoing collaboration. In developing this study, we experienced a need to schedule such meetings more frequently than we would have in collaborations staying within disciplines.
- Go into the collaboration with an open mind and a willingness to compromise. Our choice to “move” gender

involved a willingness to rethink the experiment that demonstrated exactly this. The choice accompanying the move of gender—to design a multi-method study consisting both of quantitative and qualitative methods—was also a key compromise.

- Think of the outcome of the interdisciplinary collaboration not only as tangible results to an experiment but also as a working conversation. In the study at hand this was attempted by making sure to not only discuss the experiment itself and the expected outcomes in our meetings, but also to reflect on the overarching experience of working together.

D. Limitations

We are mindful that this is our first attempt at an interdisciplinary study of gender in HRI, and that it is not without flaws. First, the sample size was relatively small and the study was conducted in English, despite we had participants from diverse linguistic and cultural backgrounds. English, like many other languages, includes traces of patriarchal social structures, with sexist attitudes embedded within its linguistic structure [33]. This may have influenced participants’ perceptions and responses.

Second, in the study we used only one type of robot (Pepper). Gender expressions may vary across different robots. This remains to be explored. Future research will aim to investigate other robots and different scenarios regarding to better understand how gender impressions emerge from the interaction.

Third, advancements in artificial intelligence and its integration into robotics could reshape gender roles in various industrial contexts. This raises important questions, such as whether robots will be perceived as having a gender, the societal implications of such perceptions, and whether robots will be expected to conform to gendered roles based on traditional human expectations (e.g., a female nurse robot). These ethical and societal considerations remain to be explored.

There are many questions remain unanswered, some of which are brought up above, that we encourage other researchers to reflect upon in other interdisciplinary discussions on gender in HRI. We hope that these suggestions will help other researchers, in HRI and elsewhere, not to fear staying with the trouble of gender [20], [34], and by doing so, continue thinking of new ways to understand and study gender in HRI.

VI. CONCLUSION

In this paper, we have contributed to the idea that successful HRI requires interdisciplinary collaboration, with a particular focus on HRI studies concerning gender. We have rooted this argument in our own experiences of collaborating across disciplinary lines, around gender and robotic healthcare providers. We have illustrated how, in this case, our discussions around the nature of gender led us to design a multi-method study building on a view of gender as fluid and performative. Finally, to encourage others to engage in interdisciplinary collaborations and discussions around gender in HRI, we have

put forth practical tips of how to carry out such collaborations. We hope that these tips will enable others to keep thinking about new ways of doing gender in HRI.

REFERENCES

- [1] K. Winkle, G. I. Melsión, D. McMillan, and I. Leite, “Boosting robot credibility and challenging gender norms in responding to abusive behaviour: A case for feminist robots,” in *Companion of the 2021 ACM/IEEE International Conference on Human-Robot Interaction*, 2021, pp. 29–37.
- [2] D. J. Rea, Y. Wang, and J. E. Young, “Check your stereotypes at the door: an analysis of gender typecasts in social human-robot interaction,” in *International Conference on Social Robotics*. Springer, 2015, pp. 554–563.
- [3] M. Paetzel, C. Peters, I. Nyström, and G. Castellano, “Congruency matters-how ambiguous gender cues increase a robot’s uncanniness,” in *International conference on social robotics*. Springer, 2016, pp. 402–412.
- [4] S. Alesich and M. Rigby, “Gendered robots: Implications for our humanoid future,” *IEEE Technology and Society Magazine*, vol. 36, no. 2, pp. 50–59, 2017.
- [5] T. Nomura, “Robots and gender,” *Gender and the Genome*, vol. 1, no. 1, pp. 18–25, 2017.
- [6] A. Pillinger, “Literature review: Gender and robotics,” 2019.
- [7] R. A. Søråa, “Mechanical genders: how do humans gender robots?” *Gender, Technology and Development*, vol. 21, no. 1-2, pp. 99–115, 2017.
- [8] J. Robertson, “Gendering humanoid robots: Robo-sexism in japan,” *Body & Society*, vol. 16, no. 2, pp. 1–36, 2010.
- [9] F. Eyssel and F. Hegel, “(s) he’s got the look: Gender stereotyping of robots 1,” *Journal of Applied Social Psychology*, vol. 42, no. 9, pp. 2213–2230, 2012.
- [10] J. Bernotat, F. Eyssel, and J. Sachse, “The (fe) male robot: how robot body shape impacts first impressions and trust towards robots,” *International Journal of Social Robotics*, vol. 13, no. 3, pp. 477–489, 2021.
- [11] S. Thellman, W. Hagman, E. Jonsson, L. Nilsson, E. Samuelsson, C. Simonsson, J. Skönvall, A. Westin, and A. Silvervarg, “He is not more persuasive than her: No gender biases toward robots giving speeches,” in *Proceedings of the 18th International Conference on Intelligent Virtual Agents*, 2018, pp. 327–328.
- [12] M. Siegel, C. Breazeal, and M. I. Norton, “Persuasive robotics: The influence of robot gender on human behavior,” in *2009 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, 2009, pp. 2563–2568.
- [13] T. Berker and A. K. Kvellheim, “Boundary objects as facilitators in sustainable building research,” *Science and Public Policy*, vol. 45, no. 2, pp. 202–210, 2018.
- [14] C. Pudlinski, “Colloquium: Dilemmas of academic discourse,” 2000.
- [15] D. Kragic and Y. Sandamirskaya, “Effective and natural human-robot interaction requires multidisciplinary research,” 2021.
- [16] P. Treusch, A. Berger, and D. K. Rosner, “Useful uselessness? teaching robots to knit with humans,” in *Proceedings of the 2020 ACM designing interactive systems conference*, 2020, pp. 193–203.
- [17] D. Bryant, J. Borenstein, and A. Howard, “Why should we gender? the effect of robot gendering and occupational stereotypes on human trust and perceived competency,” in *Proceedings of the 2020 ACM/IEEE International Conference on Human-Robot Interaction*, 2020, pp. 13–21.
- [18] B. Tay, Y. Jung, and T. Park, “When stereotypes meet robots: the double-edge sword of robot gender and personality in human–robot interaction,” *Computers in Human Behavior*, vol. 38, pp. 75–84, 2014.
- [19] K. E. Schaefer, “Measuring trust in human robot interactions: Development of the “trust perception scale-hri,” in *Robust Intelligence and Trust in Autonomous Systems*. Springer, 2016, pp. 191–218.
- [20] J. Butler, *Gender trouble*. routledge, 2002.
- [21] I. M. Young, “Throwing like a girl: A phenomenology of feminine body comportment motility and spatiality,” *Human studies*, vol. 3, no. 1, pp. 137–156, 1980.
- [22] J. Butler *et al.*, *Undoing gender*. Psychology Press, 2004.
- [23] F. Ferrando, “Is the post-human a post-woman? cyborgs, robots, artificial intelligence and the futures of gender: a case study,” *European Journal of Futures Research*, vol. 2, no. 1, p. 43, 2014.
- [24] D. Haraway, “A cyborg manifesto: Science, technology, and socialist-feminism in the late 20th century,” in *The international handbook of virtual learning environments*. Springer, 2006, pp. 117–158.
- [25] C. A. Davies, *Reflexive ethnography: A guide to researching selves and others*. Routledge, 2012.
- [26] M. Shildrick, *Embodying the monster: Encounters with the vulnerable self*. Sage, 2002.
- [27] D. Leder, *The absent body*. University of Chicago Press, 1990.
- [28] K. Winkle, D. McMillan, M. Arnelid, K. Harrison, M. Balaam, E. Johnson, and I. Leite, “Feminist human-robot interaction: Disentangling power, principles and practice for better, more ethical hri,” in *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction*, 2023, pp. 72–82.
- [29] P. Ekman and E. L. Rosenberg, *What the face reveals: Basic and applied studies of spontaneous expression using the Facial Action Coding System (FACS)*. Oxford University Press, USA, 1997.
- [30] P. Ekman, “Facial action coding system,” 1977.
- [31] S. Ahmed, *The cultural politics of emotion*. Routledge, 2013.
- [32] A. Pellegrini and J. Puar, “Affect,” *Social Text*, vol. 27, no. 3, pp. 35–38, 2009.
- [33] G. Torusdağ and R. A. Tunç, “The reflections of patriarchal culture in english: Linguistic sexism,” *Research in Education and Social Sciences*, p. 133, 2020.
- [34] D. J. Haraway, *Staying with the Trouble*. Duke University Press, 2016.